

1.

The screenshot shows the SIMATS Online Programming Lab interface. The left sidebar contains a 'QUESTIONS' section with a dropdown menu set to 'Logger'. Below it, a description states: 'You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error".' The main 'PROGRAM EDITOR' area displays the following Java code:

```
1 interface Logger {
2     void debug(String message);
3     void info(String message);
4     void warning(String message);
5     void error(String message);
6 }
7 class Log4jLogger implements Logger {
8     private String className;
9     public Log4jLogger(String className) {
10         this.className = className;
11     }
12     public void debug(String message) {
13         System.out.println("DEBUG [" + className + "] - " + message);
14     }
15     public void info(String message) {
16         System.out.println("INFO [" + className + "] - " + message);
17     }
18     public void warning(String message) {
19         System.out.println("WARN [" + className + "] - " + message);
20     }
21     public void error(String message) {
```

The right sidebar shows an 'INPUT' field with the text 'Your INPUT goes here!'. The bottom status bar indicates '37°C Mostly cloudy' and the time '14:25 30-07-2026'.

The screenshot shows the SIMATS Online Programming Lab interface. The left sidebar is the same as the previous screenshot. The main 'PROGRAM EDITOR' area displays the following Java code:

```
11     }
12     public void debug(String message) {
13         System.out.println("DEBUG [" + className + "] - " + message);
14     }
15     public void info(String message) {
16         System.out.println("INFO [" + className + "] - " + message);
17     }
18     public void warning(String message) {
19         System.out.println("WARN [" + className + "] - " + message);
20     }
21     public void error(String message) {
22         System.out.println("ERROR [" + className + "] - " + message);
23     }
24     public static void main(String[] args) {
25         Logger logger = new Log4jLogger("MainApp");
26         logger.debug("This is a debug message");
27         logger.info("This is an info message");
28         logger.warning("This is a warning message");
29         logger.error("This is an error message");
30     }
31 }
```

The right sidebar shows an 'INPUT' field with the text 'Your INPUT goes here!'. The bottom status bar indicates '37°C Mostly cloudy' and the time '14:25 30-07-2026'.

The screenshot shows the SIMATS Online Programming Lab interface. The left sidebar is the same as the previous screenshots. The main 'PROGRAM EDITOR' area displays the following Java code:

```
24     public static void main(String[] args) {
25         Logger logger = new Log4jLogger("MainApp");
26         logger.debug("This is a debug message");
27         logger.info("This is an info message");
28         logger.warning("This is a warning message");
29         logger.error("This is an error message");
30     }
31 }
```

The right sidebar shows an 'INPUT' field with the text 'Your INPUT goes here!'. The bottom status bar indicates '37°C Mostly cloudy' and the time '14:25 30-07-2026'.

4.

The screenshot shows the SIMATS Online Programming Lab interface. The left sidebar has a dropdown menu with 'HashSet' selected. Below it, the text reads: 'Write a Java program to create a HashSet of strings and remove a specific element from it.' The main area is a code editor with the following Java code:

```
1 import java.util.*;
2 public class HashSet {
3     public static void main(String[] args) {
4         HashSet<String> set = new HashSet<>();
5         set.add("Apple");
6         set.add("Banana");
7         set.add("Mango");
8         set.add("Grapes");
9         System.out.println("Before removal: " + set);
10        String removeElement = "Banana";
11        boolean removed = set.remove(removeElement);
12        System.out.println("Removed \" + removeElement + "\"? " + removed);
13        System.out.println("After removal: " + set);
14    }
15 }
```

Below the code editor is an 'OUTPUT' section that displays the error message: 'Security Error: Disallowed code pattern detected.' On the right, there is an 'INPUT' section with the text: 'Your INPUT goes here!'

7.

The screenshot shows the SIMATS Online Programming Lab interface. The left sidebar has a dropdown menu with 'LinkedHashMap' selected. Below it, the text reads: 'Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.' The main area is a code editor with the following Java code:

```
1 import java.util.LinkedHashMap;
2 public class LinkedHashMapDemo {
3     public static void main(String[] args){
4         LinkedHashMap<Integer,String>map=new LinkedHashMap<>();
5         map.put(101,"srikanth");
6         map.put(100,"trish");
7         map.put(102,"sankar");
8         map.put(99,"sri");
9         System.out.println("keys in insertion order:");
10        for(Integer key : map.keySet()){
11            System.out.println(key);
12        }
13    }
14 }
```

Below the code editor is an 'OUTPUT' section that displays the error message: 'Security Error: Disallowed code pattern detected.' On the right, there is an 'INPUT' section with the text: 'Your INPUT goes here!'

The screenshot shows the SIMATS Online Programming Lab interface. The left sidebar has a dropdown menu with 'LinkedHashMap' selected. Below it, the text reads: 'Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.' The main area is a code editor with the following Java code:

```
14 }
```

Below the code editor is an 'OUTPUT' section that displays the output of the program:

```
keys in insertion order:
101
100
102
99
```

On the right, there is an 'INPUT' section with the text: 'Your INPUT goes here!'

5.

The screenshot shows the SIMATS Online Programming Lab interface. The browser tabs include ChatGPT, Smallest Element in Array, Java Singleton Pattern | Hack..., JAVA /assessments at main..., SIMATS Online Lab, and Ask Gemini. The URL is simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=collections. The page header displays the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the subject "JAVA PROGRAMMING", and the user name "CHUNDURI NAGA SAI SRIKANTH" with ID "192311159RD".

On the left sidebar, the "TreeMap" topic is selected. The instructions state: "Write a Java program to create a TreeMap of strings and integers and iterate over its entries." The main code editor contains the following Java code:

```
1 import java.util.*;
2 public class TREEMAP{
3     public static void main(String[] args){
4         TreeMap<Integer,String>map=new TreeMap<>();
5         map.put(100,"john");
6         map.put(101,"sri");
7         map.put(102,"harsha");
8         System.out.println(map);
9     }
10 }
```

The output section shows the result: `{100=john, 101=sri, 102=harsha}`. The right sidebar has a placeholder text: "Your INPUT goes here!". The Windows taskbar at the bottom shows the date and time as 14:30 on 30-07-2026.

8.

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On the left sidebar, the "Max-Min" topic is selected. The instructions state: "Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values." The main code editor contains the following Java code:

```
1 import java.util.Set;
2 import java.util.TreeSet;
3 public class MaxMinSet {
4     public static void main(String[] args) {
5         Set<Integer> numbers = new TreeSet<>();
6         numbers.add(10);
7         numbers.add(3);
8         numbers.add(25);
9         numbers.add(-5);
10        numbers.add(17);
11        int max = ((TreeSet<Integer>) numbers).last();
12        int min = ((TreeSet<Integer>) numbers).first();
13        System.out.println("Maximum value: " + max);
14        System.out.println("Minimum value: " + min);
15    }
16 }
```

The output section shows the result: `Maximum value: 25` and `Minimum value: -5`. The right sidebar has a placeholder text: "Your INPUT goes here!". The Windows taskbar at the bottom shows the date and time as 14:31 on 30-07-2026.

6.

ChatGPT x Smallest Element in Array x Java Singleton Pattern | Hack x JAVA -assessments at main x SIMATS Online Lab x Ask Gemini

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PRIORITYQUEUE

Write a Java program to create a PriorityQueue of integers and add elements to it.

```
1 import java.util.PriorityQueue;
2 public class Main {
3     public static void main(String[] args) {
4         PriorityQueue<Integer> pq = new PriorityQueue<>();
5         pq.add(10);
6         pq.add(5);
7         pq.add(20);
8         pq.add(1);
9         pq.add(15);
10        System.out.println("PriorityQueue elements in order:");
11        while (!pq.isEmpty()) {
12            System.out.print(pq.poll() + " ");
13        }
14    }
15 }
```

OUTPUT

PriorityQueue elements in order:
1 5 10 15 20

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2.

ChatGPT x Smallest Element in Array x Java Singleton Pattern | Hack x JAVA -assessments at main x SIMATS Online Lab x Ask Gemini

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QUESTIONS 7 / 8 completed

Character

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement

PROGRAM EDITOR

Java Run Save

```
1 interface Character {
2     void move();
3     void attack();
4     void takeDamage(int damage);
5 }
6 class Hero implements Character {
7     private int health;
8     public Hero(int health) {
9         this.health = health;
10    }
11    public void move() {
12        System.out.println("Hero is moving...");
13    }
14    public void attack() {
15        System.out.println("Hero attacks with courage!");
16    }
17    public void takeDamage(int damage) {
18        health -= damage;
19        System.out.println("Hero takes " + damage + " damage. Health left: " + health);
20    }
21 }
```

INPUT

Your INPUT goes here!

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ChatGPT x Smallest Element in Array x Java Singleton Pattern | Hack x JAVA-/assessments at main x SIMATS Online Lab x Ask Gemini

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QUESTIONS 7 / 8 completed

Character

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PROGRAM EDITOR Java Run Save

```

21 }
22 class Villain implements Character {
23     private int health;
24     public Villain(int health) {
25         this.health = health;
26     }
27     public void move() {
28         System.out.println("Villain is creeping...");
29     }
30     public void attack() {
31         System.out.println("Villain attacks with evil plans!");
32     }
33     public void takeDamage(int damage) {
34         health -= damage;
35         System.out.println("Villain takes " + damage + " damage. Health left: " + health);
36     }
37 }
38 class Monster implements Character {
39     private int health;
40     public Monster(int health) {

```

INPUT

Your INPUT goes here!

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ChatGPT x Smallest Element in Array x Java Singleton Pattern | Hack x JAVA-/assessments at main x SIMATS Online Lab x Ask Gemini

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QUESTIONS 7 / 8 completed

Character

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement

PROGRAM EDITOR Java Run Save

```

37 }
38 class Monster implements Character {
39     private int health;
40     public Monster(int health) {
41         this.health = health;
42     }
43     public void move() {
44         System.out.println("Monster is stomping...");
45     }
46     public void attack() {
47         System.out.println("Monster attacks ferociously!");
48     }
49     public void takeDamage(int damage) {
50         health -= damage;
51         System.out.println("Monster takes " + damage + " damage. Health left: " + health);
52     }
53 }
54 public class main {
55     public static void main(String[] args) {
56         Character hero = new Hero(100);
57         Character villain = new Villain(100);

```

INPUT

Your INPUT goes here!

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3.

ChatGPT

Smallest Element in Array

Java Singleton Pattern | Hack

JAVA-/assessments at main

SIMATS Online Lab

Ask Gemini

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CHUNDURI NAGA SAI SRIKANTH
192311159RD

Sort

SORT

Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         List<Integer> numbers = new ArrayList<>();
5         numbers.add(5);
6         numbers.add(2);
7         numbers.add(9);
8         numbers.add(1);
9         numbers.add(7);
10        Collections.sort(numbers);
11        System.out.println("Sorted List: " + numbers);
12    }
13 }
```

OUTPUT

Sorted List: [1, 2, 5, 7, 9]

Your INPUT goes here!

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Search



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